



Influence of shrinkage on convective drying of fresh vegetables: A theoretical model



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ABSTRACT

The aim of the present work was the formulation of a theoretical model predicting the behavior of a convective drier over a wide range of process conditions. The proposed approach was based on the coupling of a transport phenomena model, describing the simultaneous transfer of momentum, heat and mass both in the drying chamber and in the food, and of a structural mechanics model aimed at estimating food sample deformations, as due to moisture loss. The effects of food shrinkage on drying performance were ascertained by analyzing the spatial distributions of temperature, moisture content, strain and stress, as a function of operating conditions. The agreement between model predictions and a set of experimental data collected during drying of cylindrical potatoes was good as far as the time evolutions of food average moisture content and of its main dimensions, i.e. length and diameter, were concerned.

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1. Introduction

Convective drying is characterized by the simultaneous transfer of momentum, heat and water, whose removal determines significant modifications to food structure. On modeling food convective drying it is essential to develop reliable simulation tools capable of predicting the actual influence of operating conditions either on dried foods characteristics or on process performance. Within this frame, multiphysics modeling plays a key role, since it combines multiple physical phenomena in a single computational environment, which may allow predicting the mutual interactions between transport phenomena, governing food drying process, and the modifications occurring in food structure.

The presence of both liquid water and vapor in the solid matrix, the dependence of food transport properties on the local values of temperature and moisture content, the complex fluid–structure interactions determined by air-flow around the sample, the variation, with time, of food shape and of its dimensions do actually make drying modeling rather problematic (Chen, 2007). Datta's research group proposed several comprehensive and general multiphase models predicting the heat and mass transfer rates in different industrial processes involving foods (Datta, 2007; Halder et al., 2011; Zhang and Datta, 2004; Ni et al., 1999; Halder et al., 2007; Zhang et al., 2005; Dhall and Datta, 2011). In these papers,

the transport phenomena occurring at food/air interfaces were described in terms of heat and mass transfer coefficients estimated from semi-empirical correlations and referred to samples having a constant characteristic dimension. When food shape is not regular or it even changes with time, i.e. when shrinkage is significant, the exploitation of literature correlations might significantly limit the model accuracy (Bernstein and Noreña, 2013), thus providing unreliable predictions of the actual system behavior. Curcio et al. (2008) formulated a theoretical model describing food convective drying without resorting to any empirical transport coefficient. The model, however, was based on several assumptions; among these, the exploitation of an effective diffusion coefficient, which did not make any distinction between the transport of liquid water and that of vapor, and the negligibility of water evaporation inside the food were definitely the most critical. In a subsequent paper, Curcio (2010) improved the predictions of the previous model and simulated drying behavior when inner evaporation could not be neglected; a multiphase approach, based on the conservation of both liquid water and vapor, was formulated. In both the above-described studies, however, food shrinkage was neglected assuming that, in the chosen range of operating conditions, food shape and its dimensions did not change appreciably as drying proceeded.

The removal of water from a porous solid material is actually responsible for the development of a field of contracting stresses in the matrix (Kowalski and Mielniczuk, 2006; Mihoubi and Bellagi, 2009; Kowalski and Rajewska, 2002). An ever-different

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Nomenclature

C^*	universal constant for smooth walls (–)	Greek symbols	
C_2	water concentration in air (mol/m ³)	α	constant comparing in $k-\omega$ model (–)
C_{pa}	air specific heat (J/(kg K))	β	constant comparing in $k-\omega$ model (–)
C_{ps}	food specific heat (J/(kg K))	β_κ	constant comparing in $k-\omega$ model (–)
C_v	vapor concentration in food (mol/m ³)	δ_w	distance from the wall (m)
C_w	liquid water concentration in food (mol/m ³)	$d\varepsilon$	local total strain (–)
D	potato diameter (m)	$d\varepsilon_0$	shrinkage strain (–)
D_a	diffusion coefficient of water in air (m ² /s)	$d\varepsilon_{r0}$	shrinkage strain in radial direction (–)
dC_w	water concentration variation (mol/m ³)	$d\varepsilon_s$	mechanical strain (–)
D_v	effective diffusion coefficient of vapor in food (m ² /s)	$d\varepsilon_{z0}$	shrinkage strain in axial direction (–)
D_w	capillary diffusion coefficient of water in food (m ² /s)	η_a	air viscosity (Pa s)
E_y	Young modulus (Pa)	$\eta_t = \rho_a k / \omega$	air turbulent viscosity (Pa s)
$G_m = E_y / [2 * (1 + \nu)]$	shear modulus (Pa)	κ	von Karman's constant (–)
H'	strain-hardening rate (Pa)	λ	water latent heat of vaporization (J/mol)
$\dot{I}$	volumetric rate of evaporation (mol/(m ³ s))	ν	Poisson ratio (–)
k	turbulent kinetic energy (m ² /s ²)	ξ	yield stress parameter
k_a	air thermal conductivity (W/(m K))	ρ_a	air density (kg/m ³)
K_{ds}	constant in Eq. (5) (m ³ /mol)	ρ_s	food density (kg/m ³)
k_{eff}	effective thermal conductivity of food (W/(m K))	σ	stress (Pa)
L	potato length (m)	σ'	deviatoric stress (Pa)
$\underline{n}$	unity vector normal to the surface (–)	$\bar{\sigma}$	equivalent stress (Pa)
p	pressure within the drying chamber (Pa)	σ_d	yield stress (Pa)
p_v	vapor pressure of water (Pa)	σ_k	constant comparing in $k-\omega$ model (–)
p_{vs}	saturated vapor pressure of water (Pa)	σ_ω	constant comparing in $k-\omega$ model (–)
r	radial coordinate (m)	ω	dissipation per unit of turbulent kinetic energy
T	food temperature (K)		
t	time (s)	Subscripts	
T_2	air temperature (K)	0	initial condition ($t = 0$)
T_{air}	air temperature at the drier inlet (K)	atm	atmospheric conditions
$\underline{u}$	averaged velocity field (m/s)	r	in radial direction
$\underline{u}'$	fluctuating part of velocity field (m/s)	rz	referred to shear
U_r	air relative humidity at the drier inlet (–)	z	in axial direction
u_τ	friction velocity (m/s)	θ	angular direction
V	food volume (m ³)		
v_0	air velocity at the drier inlet (m/s)	Superscripts	
X_b	moisture content on a dry basis kg water/kg dry solid	I	referred to first principal stress
$\bar{X}_b$	average moisture content on a dry basis kg water/kg dry solid		
z	axial coordinate (m)		

mechanical equilibrium of the material is attained and a change both of its shape and dimensions is observed (Mayor and Sereno, 2004; Aregawi et al., 2012). Drying methods as well as the exploited operating conditions differently affect both the quality and the main characteristics of dried foods, including volume and shape changes (Panyawong and Devahastin, 2007). In the case of vegetables undergoing drying under different conditions, Ratti (1994) proposed that shrinkage characteristics were strictly dependent on food moisture content; in addition, the observed variations in the surface-to-volume ratio were a function of sample geometry and of the type of foodstuff. Yadollahinia and Jahangiri (2009) analyzed the influence of various drying temperatures and air velocities on the shrinkage of potato slices undergoing convective drying. Ramos et al. (2004) proposed the exploitation of a microstructural approach to quantify the physical changes occurring during air-drying of grapes quarters, so to ascertain the actual cellular shrinkage. Hassini et al. (2007) reported that shrinkage effect could not be neglected when moisture diffusivity in highly shrinking materials, like vegetables and fruits, had to be determined. The extent of shrinkage strongly depends on matrix mobility; in particular, it was proved that shrinkage is more significant during the constant and the falling rate periods, i.e.

when matrix mobility is higher (Karathanos, 1993; Rahman, 2001; Shishegarha et al., 2002; Troncoso and Pedreschi, 2007). Katekawa and Silva (2007), observed that the reduction of food volume actually corresponded to the volumetric amount of liquid water removed from the sample (ideal shrinkage).

A comprehensive mathematical model describing shrinkage phenomenon of materials undergoing drying processes was firstly proposed by Kowalski (1996). The model was based on the methods of continuum mechanics and on the principles of thermodynamics of irreversible processes. Hernandez et al. (2000) provided a mathematical description of food drying kinetics taking into account the effect of shrinkage. The authors validated the model consistency by performing various experiments under different drying conditions for fruits available in two shapes. Several empirical models were also formulated to fit the experimental results collected during drying of different foods and expressing the variation of samples volume vs. its moisture content (Aversa et al., 2012). Mayor and Sereno (2004) summarized the results obtained by different authors and concluded that shrinkage affects the predictions both of moisture and temperature profiles. Shrinkage, therefore, has necessarily to be taken into account when a mathematical model aimed at describing drying process is to be formulated (Márquez

and De Michelis, 2011). Yang et al. (2001) proposed a model, based on the coupling of transport equations and of the virtual work principle, for the prediction of heat and moisture transfer occurring during drying of a cylindrical potato sample. The model neglected the transport phenomena in the air and exploited a set of heat and mass transfer coefficients estimated from the empirical correlations available in the literature; in addition, it was assumed that strain displacement was proportional to food shrinkage coefficient. The latter assumption was also used by Niamnuy et al. (2008), who proposed a drying model, which did not consider the transport phenomena occurring in the drying air, too.

The present work was aimed at formulating a multiphase transport model predicting the influence of operating conditions on food shrinkage as well as on drying process performance. The model was based on the conservation of liquid water, vapor and energy in food, coupled to the conservation of vapor and energy in the drying air. It was assumed that the flux of water (both as liquid and vapor) reaching the food outer surfaces from the interior was fully evaporated and convected away, as vapor, by the drying air, whose velocity field strongly affected the interfacial rates of heat and mass transfer. In order to estimate food shape variations, the above-mentioned transport equations were coupled to a structural mechanics model describing the deformations occurring in the food sample as a consequence of moisture removal.

As compared to the drying models available in the literature, the main innovations introduced by the present paper regarded: (a) the description of the anisotropic change of food dimensions, as determined by the local values of moisture transfer rate. (b) The evaluation, without resorting to any semi-empirical correlation, of the interfacial heat and mass fluxes; in the case of shrinking materials, this definitely represented a significant breakthrough since the correlations currently available were obtained defining a constant characteristic geometric dimension, generally referred to regular shapes. The present model, instead, accounted for the continuous variation of food shape, as well as of the exposed surfaces, thus allowing reliable predictions of real driers performance over a wide range of process and fluid–dynamic conditions.

2. Theory

The food under study was a cylindrical potato sample having an initial length of 0.03 m and an initial diameter of 0.016 m. The potato sample was placed on a stainless steel ultrathin wide-mesh net, 0.04 m far from the inlet section of a cylindrical drying cell, whose length and diameter were equal to 0.15 m and 0.16 m, respectively (Fig. 1). The initial dimensions of fresh potato and the size of drying cell were chosen so to simulate the performance of a typical lab-scale drier and to analyze the behavior of a food sample whose geometrical characteristics were considered as representative of several dehydrated vegetables, commercialized all over the world. Drying air, characterized by definite values of temperature (T_{air}), relative humidity (U_r) and velocity (v_0) measured at drier inlet section, was continuously flowing in the axial direction, z . As described in the following, the present model was referred to spatial domains (food and air), which changed their shape and dimension because of potato anisotropic shrinkage.

The main assumptions exploited to formulate the transport model were: (a) Cylindrical symmetry, which allowed restricting the analysis to a 2D geometry. (b) Thermal equilibrium of all the phases involved in the drying process. (c) Negligibility of any possible interaction, regarding both the transport phenomena and the structural mechanics calculations, at the contact points between food and net. (d) Negligibility of gas pressure gradients for the calculation of both liquid water and vapor transport in the food; the conservation equation referred to air transport in the food was ignored as

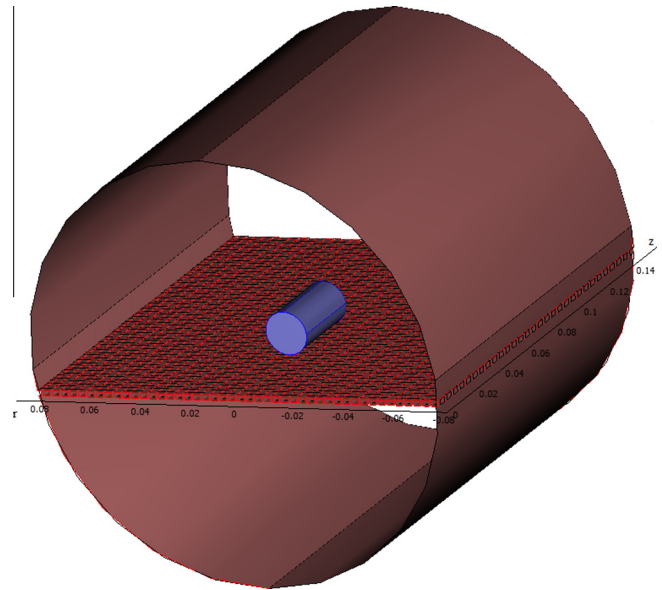


Fig. 1. Schematic representation of the drying cell.

well. (e) Negligibility of convective heat transfer in food sample. (f) The food sample was considered as a fictitious continuum, although it is actually a hygroscopic porous medium. Appendix A summarized the unsteady-state heat and mass balance equations, referred to the transport of liquid water and vapor in the food sample. It is worthwhile observing that the proposed transport model, as based on some of the previous assumptions, was suitable to describe mild-temperature drying processes, for which convective contributions could be considered as not particularly relevant.

The velocity field of drying air, flowing in turbulent conditions within the drying chamber around the potato sample, was described by the $k-\omega$ model (Wilcox, 2004), which is based on two additional semi-empirical transport equations for the turbulent kinetic energy, k , and the dissipation for unit of turbulent kinetic energy, ω , respectively. The unsteady-state momentum balance coupled to the continuity equation and to the transport equations for both k and ω were reported in Appendix B, together with the energy and the water vapor balance equations, both accounting for convective and conductive/diffusive contributions.

Appendix C summarized the set of initial and boundary conditions exploited to complete the definition of the transport model and, therefore, to carry out the numerical simulations, which were aimed at determining the effect of operating conditions, namely T_{air} , U_r and v_0 , on food shrinkage and on drying process behavior.

The structural changes occurring in potatoes as drying proceeded were estimated coupling the transport model to a structural mechanics model, which allowed calculating the stresses developing in the sample as a result of water removal. According to Yang et al. (2001), it was assumed that potatoes behavior was elastoplastic within a not large strain region. In addition, the local total strains $\{d\epsilon\}$, actually a function of the total displacements, were related to the changes both of mechanical strains $\{d\epsilon_s\}$, i.e. the constrained deformation due to mechanical elastoplasticity, and of shrinkage strains $\{d\epsilon_0\}$, representing the free deformation due to moisture loss:

$$\{d\epsilon\} = \{d\epsilon_s\} + \{d\epsilon_0\} \quad (1)$$

Following Yang et al. (2001), the stress–strain relationship was expressed through the elastoplastic stress–strain matrix, which accounted for the Young modulus, E_y , the Poisson ratio, ν , the strain-hardening rate, H' , the shear modulus, G_m , the equivalent stress, $\bar{\sigma}$, and the deviatoric stresses, σ'_r , σ'_z , σ'_θ :

$$\begin{Bmatrix} d\sigma_r \\ d\sigma_z \\ d\sigma_\theta \\ d\tau_{rz} \end{Bmatrix} = \begin{Bmatrix} \frac{E_y(1-\nu)}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1 & \nu/1-\nu & \nu/1-\nu & 0 \\ \nu/1-\nu & 1 & \nu/1-\nu & 0 \\ \nu/1-\nu & \nu/1-\nu & 1 & 0 \\ 0 & 0 & 0 & (1-2\nu)/2(1-\nu) \end{bmatrix} \begin{Bmatrix} d\epsilon_{sr} \\ d\epsilon_{sz} \\ d\epsilon_{s\theta} \\ d\gamma_{srz} \end{Bmatrix} \\ -\frac{9\zeta G_m^2}{\sigma(3G_m+H^2)} \begin{bmatrix} \sigma_r'^2 & \sigma_r'\sigma_z' & \sigma_r'\sigma_\theta' & \sigma_r'\tau_{rz}' \\ \sigma_r'\sigma_z' & \sigma_z'^2 & \sigma_z'\sigma_\theta' & \sigma_z'\tau_{rz}' \\ \sigma_r'\sigma_\theta' & \sigma_z'\sigma_\theta' & \sigma_\theta'^2 & \sigma_\theta'\tau_{rz}' \\ \sigma_r'\tau_{rz}' & \sigma_z'\tau_{rz}' & \sigma_\theta'\tau_{rz}' & \tau_{rz}^2 \end{bmatrix} \begin{Bmatrix} d\epsilon_{sr} \\ d\epsilon_{sz} \\ d\epsilon_{s\theta} \\ d\gamma_{srz} \end{Bmatrix} \end{Bmatrix} \quad (2)$$

The yield stress parameter, ζ allowed for the transition from elastic to elastoplastic deformation. In particular:

$$\zeta = 0 \text{ when } \bar{\sigma} < \sigma_d \text{ for any } \bar{\sigma}_d \bar{\sigma} \quad (3)$$

$$\zeta = 0 \text{ when } \bar{\sigma} < \sigma_d \text{ and } \bar{\sigma}_d \bar{\sigma} > 0 \quad (4)$$

where σ_d was the uniaxial yield stress.

As reported in Table 1, both E_y and σ_d were expressed in terms of the local value of food moisture content on dry basis; whereas, the Poisson ratio was assumed as constant and equal to 0.492.

To complete the formulation of the present transport/structural mechanics model, it was assumed, as reported by several papers dealing with drying of different materials (Kowalski et al., 2010; Mihoubi and Bellagi, 2009; Benboudjemaa et al., 2005), that the variation with time of the free shrinkage strain, $\{d\epsilon_0\}$, was proportional to the water concentration variation, dC_w , defined with reference to the initial water concentration in the sample:

$$\{d\epsilon_0\} = K_{ds} \cdot dC_w \quad (5)$$

As described in the following, the validity of Eq. (5) was proved by a set of experimental tests performed on potatoes sample, dried in a definite range of operating conditions.

Table 1

Young modulus, E_y , and uniaxial yield stress, σ_d for potatoes as a function of moisture content on a dry basis (Yang et al., 2001).

$E_y = 1.691 \cdot 10^7 \exp(-0.522 X)$	$X > 1.0$
$E_y = 5.23 \cdot 10^7 \exp(-1.704 X)$	$X \leq 1.0$
$\sigma_d = 1.453 \cdot 10^6 \exp(-0.164 X)$	$X > 0.75$
$\sigma_d = 7.26 \cdot 10^7 \exp(-5.04 X)$	$X \leq 0.75$

The virtual work principle was eventually formulated to obtain the equilibrium equation necessary to complete the structural mechanics model. Supposing that zero body and surface forces were applied to food, it was written as:

$$\int_V \delta\{d\epsilon_s\}^T \{d\sigma\} dV = 0 \quad (6)$$

As far as the boundary conditions were concerned, it was assumed that all the sides of food sample were free to move, except that one lying on the drier net for which a displacement in radial direction was actually precluded.

On the basis of the above discussion, it is evident that the time evolution of food moisture content had a strong influence on the total strains and, therefore, on sample displacements. These, in turn, were responsible for a variation of food shape and dimensions. Such a variation significantly affected the transport phenomena occurring at the food/air interfaces and, therefore, the actual drying rates, too. The continuous and anisotropic variation of food/air interfaces, as determined by shrinkage, was described by introducing a local modification of food and air domains. The transport equations, written for both air and food, and the virtual work principle (Eq. (6)) based on the stress–strain relationship expressed by Eq. (2) represented a system of unsteady, non-linear, partial differential equations that could be solved only by a numerical method. The Finite Elements Method (FEM), implemented in Comsol Multiphysics 3.4, was exploited. The system of PDEs was referred to a time dependent deformed mesh, accounting for food shrinkage. The variation with time of integration domains was described by an Arbitrary–Lagrangian–Eulerian (ALE) method (Hirt et al., 1997). The deformed mesh motion, strictly dependent on moisture removal and resulting from the solution of structural mechanics problem, was modeled by the Laplace smoothing. Food and air domains were discretized into a total number of 21,272 triangular finite elements leading to about 324,000° of freedom. In particular, the mesh consisted of 5328 elements (with a minimum element quality of 0.888) and of 15,944 elements (with a minimum element quality of 0.806), respectively, for food and air. The considered mesh provided an adequate spatial resolution for the system under study. The solution was independent of the grid size, even with further refinements. Lagrange finite elements of order two were chosen for the components of air velocity vector, for the turbulent

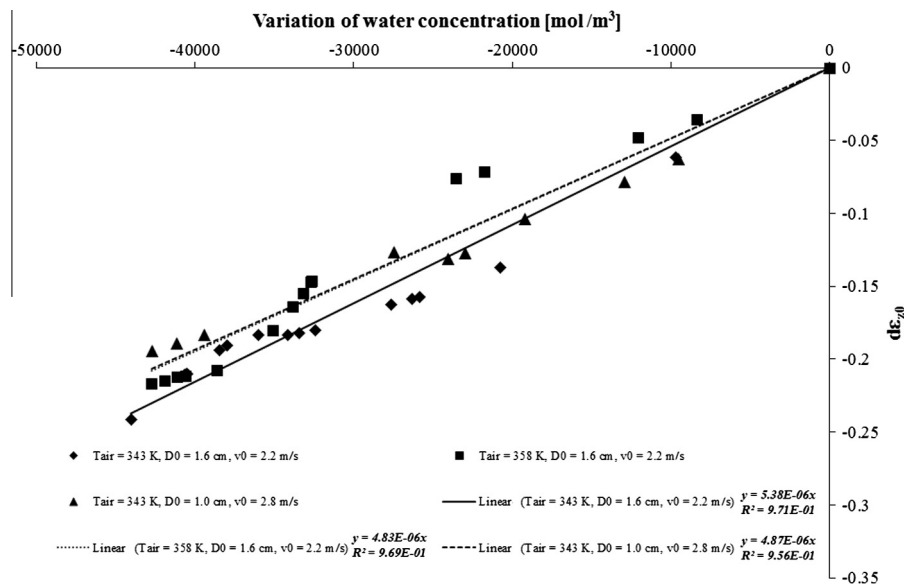


Fig. 2a. Linear fitting of $d\epsilon_0$ vs. dC_w in some of the tested experimental conditions.

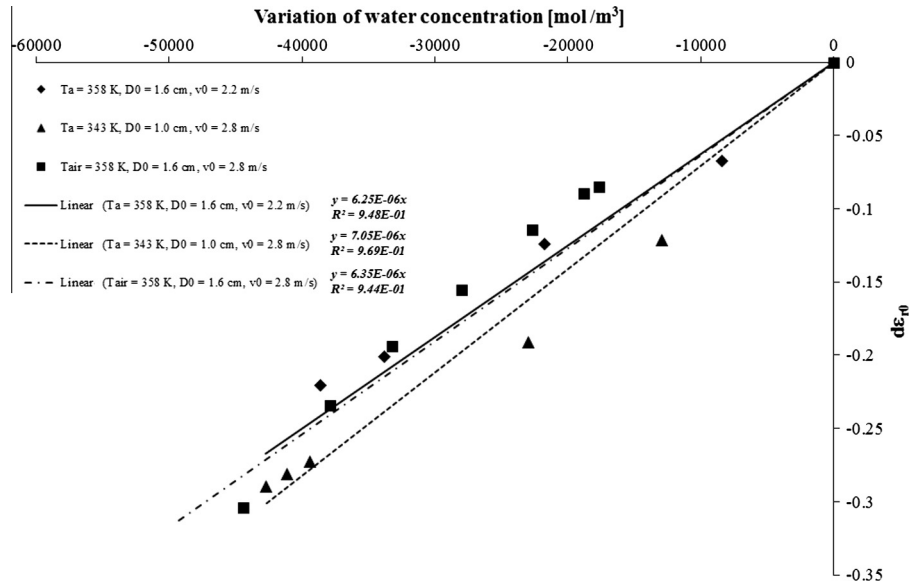


Fig. 2b. Linear fitting of d_{cr0} vs. dC_w in some of the tested experimental conditions.

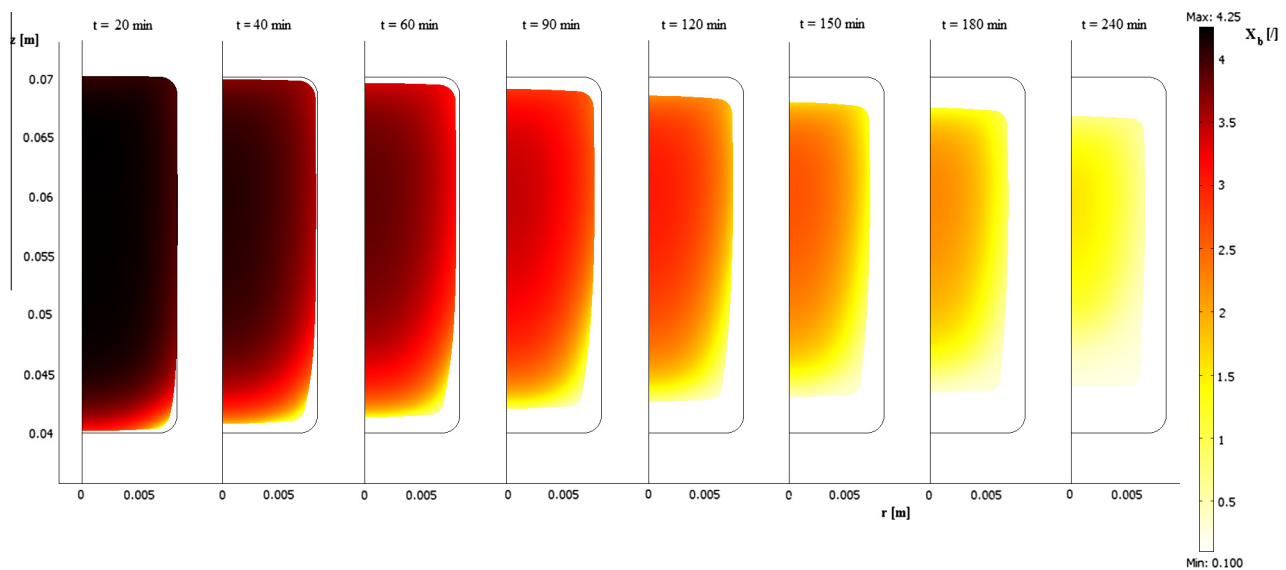


Fig. 3a. Time evolution of potato moisture content distribution ($T_{air} = 343 \text{ K}$, $U_r = 50\%$, $v_0 = 2.2 \text{ m/s}$, $X_{b0} = 4.06$).

kinetic energy, the dissipation for unit of turbulent kinetic energy and for the pressure distribution within the drying cell. Lagrange finite elements of order two were chosen, as well, for food displacements and for water concentration and temperature both in air and in food. The time-dependent problem was solved by an implicit time-stepping scheme, leading to a non-linear system of equations for each time step. The Newton's method was used to solve each non-linear system of equations, whereas a direct linear solver was adopted to solve the resulting systems of linear equations. The relative and absolute tolerances were set to 0.001 and 0.0001, respectively. On a i7 quad-core processor @ 2.67 GHz computer (12 GB DDR3 RAM) running under Windows Vista, a typical drying process lasting 5 h was simulated in about 2.5 h.

It is worthwhile remarking that the formulated model, due to its versatility and thanks to one of the most useful features of Comsol Multiphysics software, could be parameterized in order to easily change the dimensions (length and diameter) of either the drying cell or the potato sample. This allowed analyzing the effect

of a sizes variation on drying process behavior and might pave the way for the identification of some general rules aimed at properly designing the drying chamber.

3. Materials and methods

The validity of the proposed theoretical approach was verified by an experimental analysis carried out on cylindrical potatoes, dried by air of known characteristics. A standard procedure was adopted to perform the experiments, which were aimed at monitoring the time evolutions both of potato dimensions, i.e. length and diameter, and of food weight. Potatoes, purchased at the local market, were cut by a proper tool, which allowed obtaining regular cylinders having a fixed length, L , of $0.03 \pm 10^{-4} \text{ m}$ and a diameter, D , of either $0.01 \pm 10^{-4} \text{ m}$ or $0.016 \pm 10^{-4} \text{ m}$. Samples initial dimensions were measured by a vernier caliper having an accuracy of 0.02 mm. Some fragments, randomly collected from the same

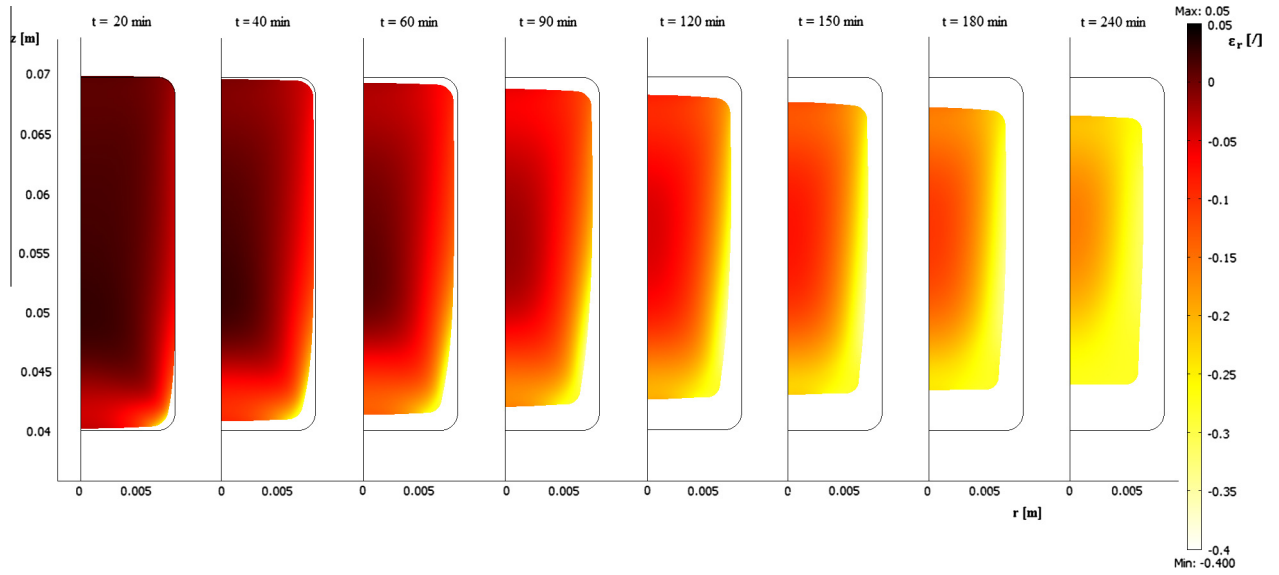


Fig. 3b. Time evolution of normal strain distribution in radial direction ($T_{air} = 343$ K, $U_r = 50\%$, $v_0 = 2.2$ m/s, $X_{b0} = 4.06$).

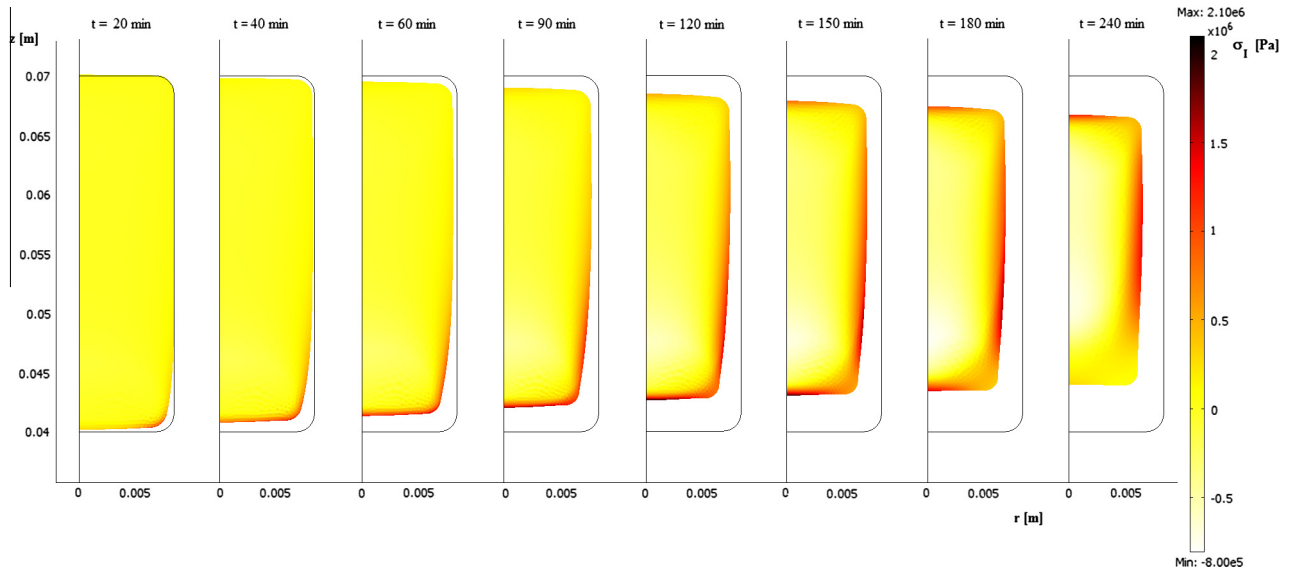


Fig. 3c. Time evolution of first principal stress distribution ($T_{air} = 343$ K, $U_r = 50\%$, $v_0 = 2.2$ m/s, $X_{b0} = 4.06$).

potato used to perform the experiments, were used to measure beforehand the initial moisture content of each sample by an electronic moisture analyzer. The average initial moisture content of the tested samples was equal to 0.805 ± 0.05 , on a wet basis. A cylindrical potato sample was placed 0.04 m far from the inlet section of a lab-scale drying cell consisting of a cylinder having a diameter of 0.16 m and a length of 0.15 m, equipped with a stainless steel ultrathin wide-mesh net. The convective flow of drying air was obtained by a line of fans placed along the edge of the internal drier tray; air entered the drier in the axial direction and flowed parallel to the food sample axis. Three different probes, produced by Dostmann electronic GmbH, Germany, measured the main characteristics of air at the drying cell inlet section. In particular, a P655 probe was used to monitor dry-bulb temperature, T_{air} , whereas a RH071073 probe and a H113828 probe allowed measuring the relative humidity, U_r , and the inlet velocity, v_0 , respectively. To achieve the validation of the present model, the experimental results collected at two air temperatures (343 K and 358 K), at

two inlet velocities (2.2 m/s and 2.8 m/s) and a relative humidity ranging from 35% to 50% will be exploited. Each experiment had a maximum duration of 5.5 h.

Weight losses, measured at given time intervals by a precision balance (Mettler AE 160, accuracy of ± 0.0001 g), allowed calculating the decay of potato average moisture content. The time evolution of both L and D were monitored, carefully removing the sample from the drying cell, and measuring the potato length and its diameter (at half of the length) by the above-described vernier caliper. Such measurements took place in less than 5 s in order to avoid any sample re-humidification.

The collected experimental data were also used to ascertain the validity of Eq. (5) and to estimate the value of K_{ds} to be exploited by the present theoretical model. The free drying shrinkage strain components $d\epsilon_{z0}$ and $d\epsilon_{r0}$ in either axial or radial direction were calculated, respectively, as:

$$d\epsilon_{z0} = (L - L_0)/L_0 \quad (7)$$

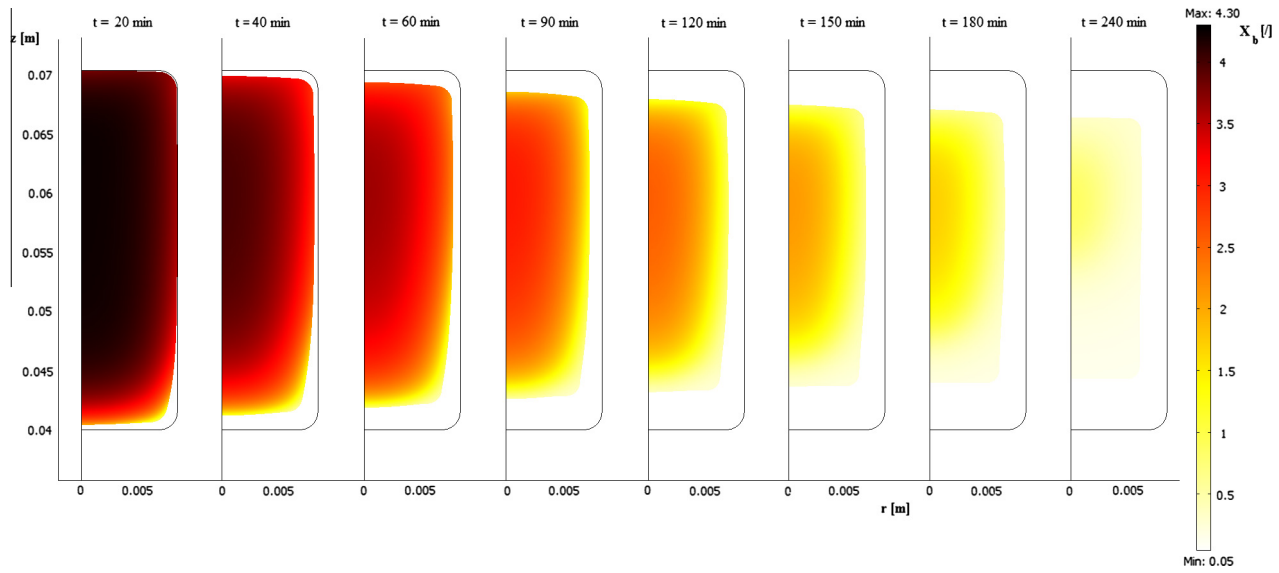


Fig. 4a. Time evolution of potato moisture content distribution ($T_{air} = 358$ K, $U_r = 35\%$, $v_0 = 2.2$ m/s, $X_{b0} = 4.06$).

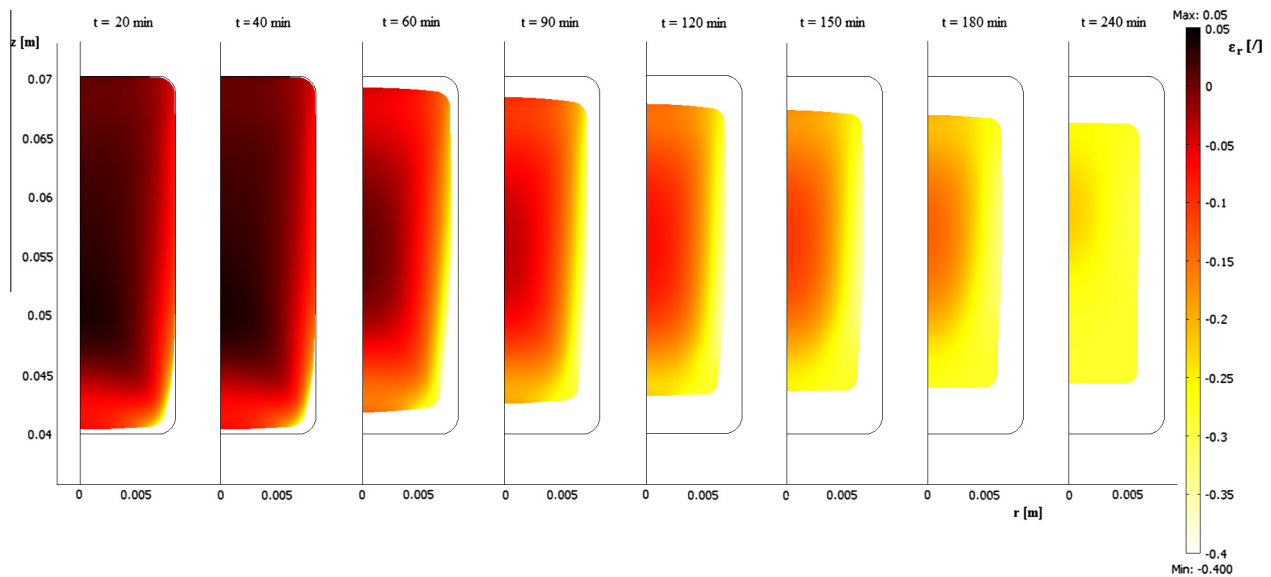


Fig. 4b. Time evolution of normal strain distribution in radial direction ($T_{air} = 358$ K, $U_r = 35\%$, $v_0 = 2.2$ m/s, $X_{b0} = 4.06$).

$$d\epsilon_{r0} = (D - D_0)/D_0 \quad (8)$$

where L_0 and D_0 indicated, respectively, the values of potato length and diameter measured at the beginning of drying process. Assuming that the decrease in food weight could be ascribed to water evaporation only, the corresponding time variation of water concentration, dC_w , was estimated on the basis of the difference between the measured value of sample weight and its initial weight.

3.1. Statistical analysis

Each experiment was repeated twice. The statistical analysis was performed using Statgraphics V 7.0 software (Statistical Graphics, Englewood Cliffs, NJ, USA) and the level of statistical significance was $p < 0.05$. The effectiveness of the proposed models

was evaluated by the coefficient of determination, R^2 , and by the percentage error between predicted and experimental points.

4. Results and discussion

Before simulating the present transport/structural mechanics model, it was necessary to prove the validity of Eq. (5). Figs. 2a and 2b presented the experimental results showing, respectively, $d\epsilon_{z0}$ and $d\epsilon_{r0}$ vs. dC_w and the corresponding linear fitting in some typical situations. The rather high values of the coefficient of determination confirmed the reliability of Eq. (5); in addition, the calculated slopes laid in a relatively limited range, thus suggesting a nearly constant value of K_{ds} , which was practically independent, at least in the chosen range, on the process conditions exploited to perform the experiments. Taking into account all the estimated

values of K_{ds} , an average value of 6.01×10^{-6} (standard deviation: 7.67×10^{-7}) was hereafter exploited to carry out the numerical simulations, aimed at predicting the behavior of potatoes convective drying.

The proposed transport/structural mechanics approach allowed achieving a detailed description of the influence of operating conditions on several variables, which definitely play a key role on drying process behavior. The formulated model does indeed represent a reliable tool for the identification of the regions (or the points) where potato moisture content, strictly related to water activity, and its temperature might determine microbial spoilage (Valdramidis et al., 2006). In addition, both the strains and the stresses developing within food sample and resulting from moisture removal may provide very useful indications about dried potatoes mechanical properties, which certainly represent important indexes ascertaining food quality. A comprehensive analysis about the mutual interactions among the operating conditions,

the distribution of some key variables, namely temperature, moisture content, strains, stresses, and the product quality and its safety could pave the way for a true optimization of vegetables drying process. Such an optimization has to be based on the definition of a proper objective function, which subjected to a set of constraints, has to be minimized (maximized) by changing the operating conditions in which drying is performed (Curcio et al., 2012).

Figs. 3a–3c showed, respectively, the time evolutions of moisture content and the corresponding normal strains in radial direction and first principal stress, developing in food sample (for sake of simplicity, only half of the sample was shown) when drying was performed in milder operating conditions. Figs. 4a–4c showed the distributions of the same variables, when drying process was carried out under more drastic conditions. In both cases, food had not an isotropic deformation. During the initial stages of drying, capillary diffusivity had large values and food moisture

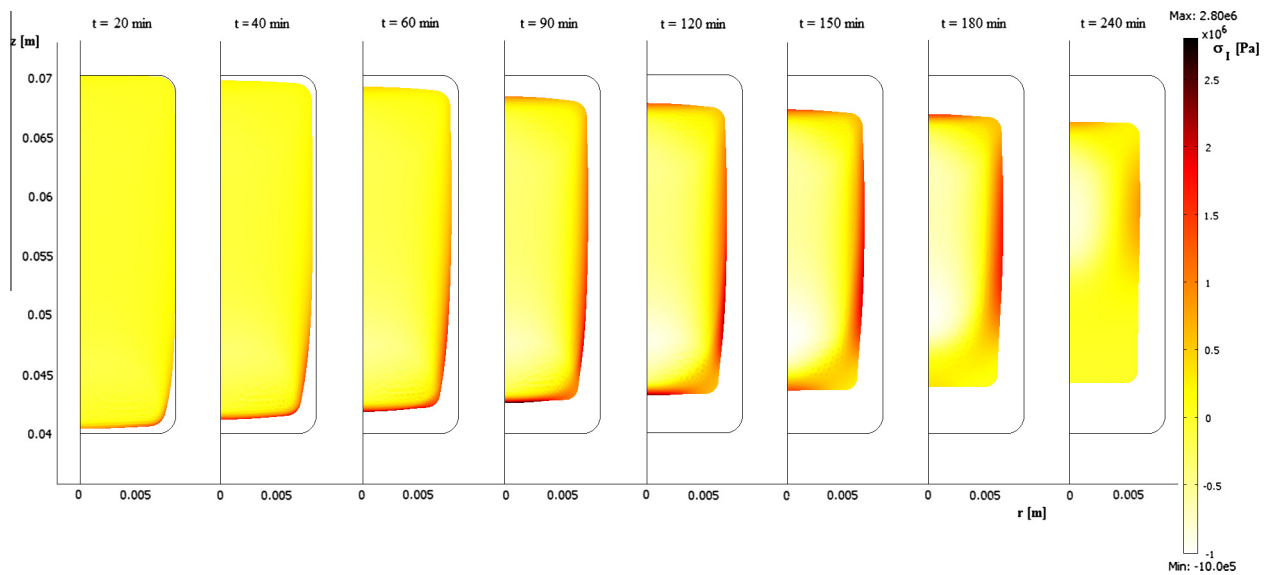


Fig. 4c. Time evolution of first principal stress distribution ($T_{air} = 358$ K, $U_r = 35\%$, $v_0 = 2.2$ m/s, $X_{b0} = 4.06$).

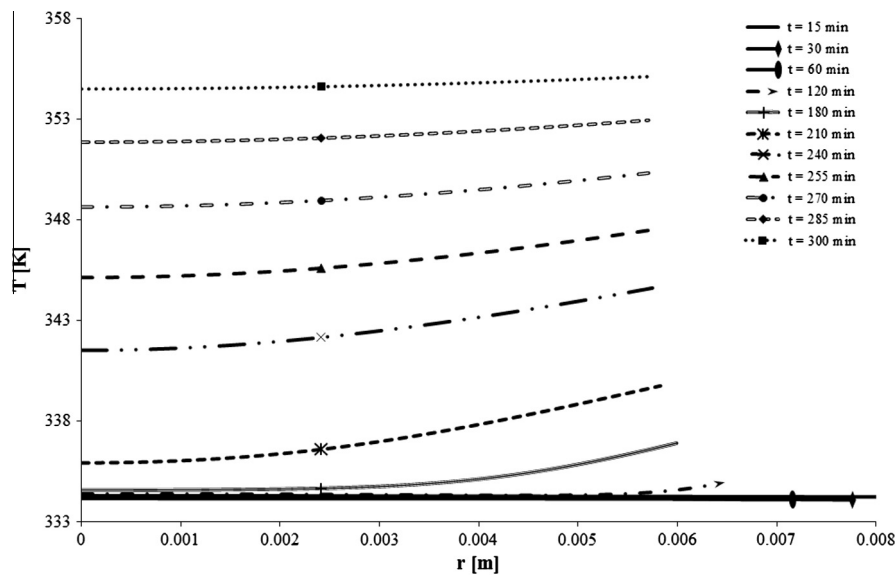


Fig. 5. Time evolution of temperature profile in radial direction ($T_{air} = 358$ K, $U_r = 35\%$, $v_0 = 2.2$ m/s, $X_{b0} = 4.06$, $z = L/2$).

content exhibited an almost uniform value, with slight differences between the core and the outer surfaces. Correspondingly, both the strain and the first principal stress distributions tended to nil, except on those sides where air actually impinged; here moisture removal was faster and a much larger (and faster) shrinkage was achieved. Such differences in moisture content tended progressively to enlarge due to both a decrease of capillary diffusivity and an increase of surface temperature, which started rising above the wet bulb temperature, as shown, at $z = L/2$, in Fig. 5. This increase in moisture gradients led to a non-uniform strain distribution and was responsible, together with the corresponding decrease of food structure mobility, for the development of either tensional stresses (plus sign) or compressive stresses (minus sign) in the food. As drying proceeded, potato surfaces attempted to shrink but were restrained by the wet core; the external surfaces and, particularly, those ones were air impinged were thus subjected to tension, while the wet core was subjected to compression.

It is worthwhile observing that extensive tensile stresses might determine a possible crack on food surface, which, therefore, could be damaged if an improper set of operating conditions was exploited to perform drying. It could be also observed that, when internal resistances to water transport controlled the drying rate (falling drying rate period), moisture content decrease led to a significant diminution of food structure mobility (Mayor and Sereno, 2004), which limited any subsequent shrinkage, thus increasing the compression stress towards the food core. Such a strong relationship among process conditions and structural mechanics variables was clearly elucidated by the present computational tool, which allowed calculating the actual numerical values of many key parameters strictly related to food quality.

Fig. 6, showing air velocity field as a function of process time, further confirmed the importance of considering food shrinkage. As drying proceeded, food dimension decreased and a progressively larger portion of its original volume was replaced by air,

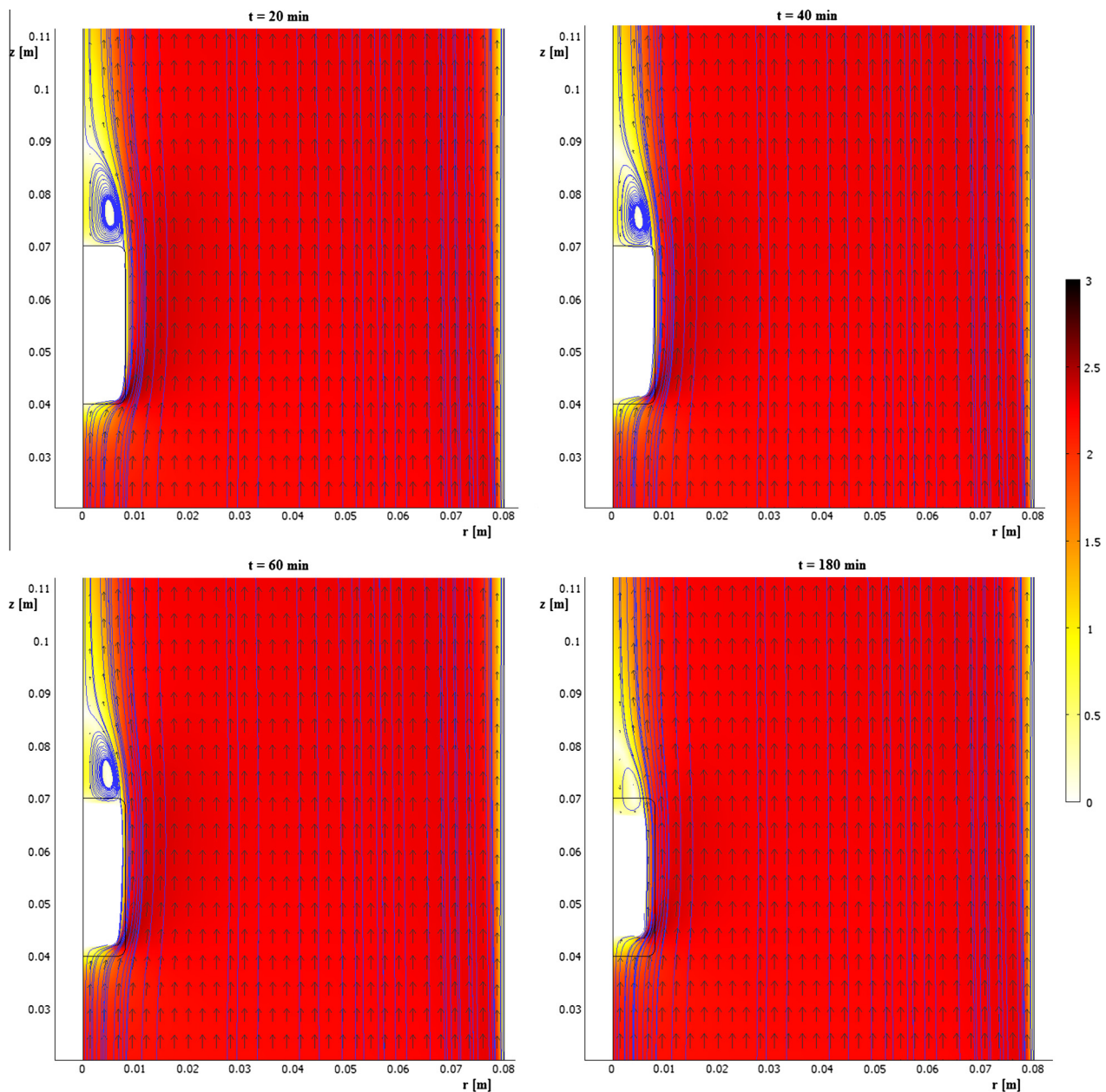


Fig. 6. Time evolution of air velocity field and of the streamlines calculated accounting for food shrinkage ($T_{air} = 358$ K, $U_r = 35\%$, $v_0 = 2.2$ m/s, $X_{b0} = 4.06$).

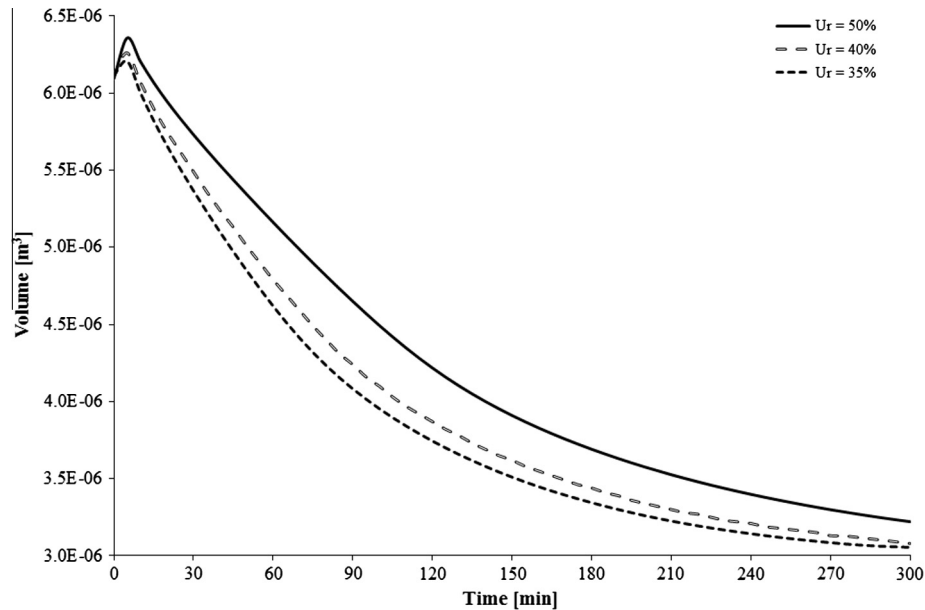


Fig. 7. Time evolution of potato volume as a function of air relative humidity ($T_{air} = 358$ K, $v_0 = 2.2$ m/s, $X_{b0} = 4.15$, $L_0 = 0.03$ m, $D_0 = 0.016$ m).

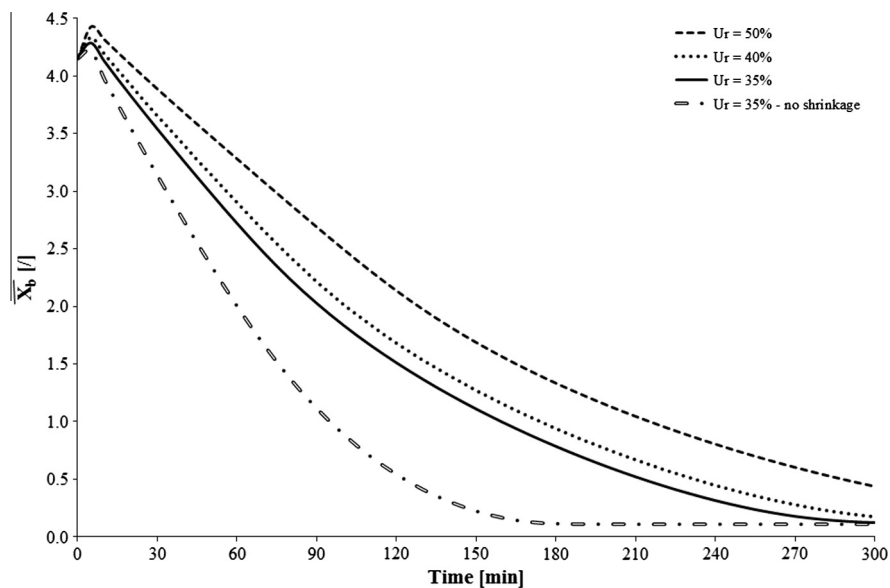


Fig. 8. Time evolution of potato average moisture content (on a dry basis) as a function of air relative humidity and comparison with the theoretical predictions obtained neglecting food shrinkage ($T_{air} = 358$ K, $v_0 = 2.2$ m/s, $X_{b0} = 4.15$, $L_0 = 0.03$ m, $D_0 = 0.016$ m).

whose streamlines, particularly in the vicinity of food exposed surfaces, were significantly changed. Such an anisotropic modification of food shape and dimensions and the corresponding variation of air velocity profiles did actually have a great influence on heat and mass transfer rates at food/air interfaces. It is worthwhile remarking that the present transport model did not rely on the exploitation of any semi-empirical correlations; rather, it was based on the formulation of a set of boundary conditions expressing the continuity of interfacial heat and mass fluxes. Therefore, the proposed approach was particularly suitable to predict the actual process performance when food geometric characteristics changed locally and depended on process time, too.

The present model could also describe the behavior of drying process from a macroscopic point of view, by analyzing the time-evolution of some characteristic quantities averaged over the

sample volume, actually changing with time due to shrinkage. Figs. 7 and 8 showed, respectively, the predicted time evolutions of both potato volume and its average moisture content on a dry basis, $\bar{X}_b$, as a function of U_r . It is worthwhile observing (Fig. 7) that during the constant drying rate period, i.e. when process rate was controlled by external resistance to heat and mass transfer, shrinkage was more pronounced and almost uniform. On the other hand, under the control of internal resistances, i.e. during the falling rate period, a less significant shrinkage was observed because of moisture content decrease, which led to a diminution of food structure mobility. As far as food average moisture content was concerned (Fig. 8), it could be observed that, initially, a slight increase in moisture content occurred, due to vapor condensation on food surfaces. During the constant rate period, drying rate attained its maximum value and water inside the food, mostly present as liquid, was

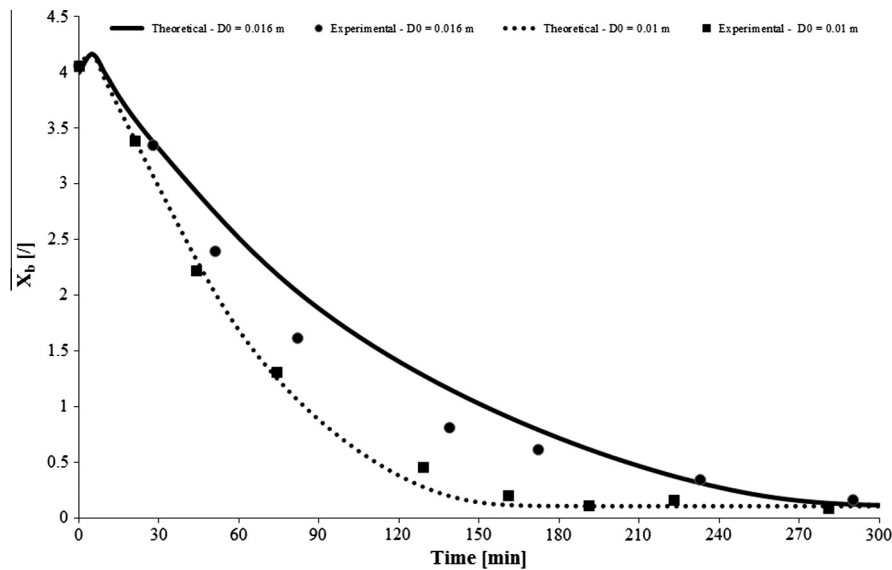


Fig. 9. Model validation – average moisture content on a dry basis vs. time ($T_{air} = 358$ K, $U_r = 35\%$; $v_0 = 2.2$ m/s; $X_{b0} = 4.06$).

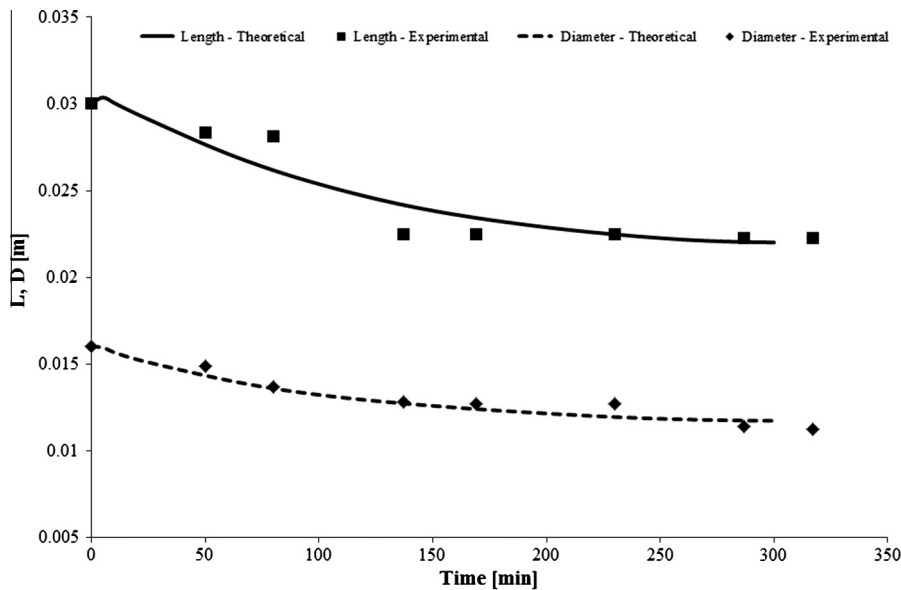


Fig. 10a. Model validation – length and diameter vs. time ($T_{air} = 358$ K, $U_r = 40\%$; $v_0 = 2.8$ m/s; $L_0 = 0.03$ m; $D_0 = 0.016$).

transferred by capillary diffusion. During the falling rate period, the weakly bound water was removed and a progressive decrease of drying rate was observed. In the considered case, only the curve corresponding to U_r equal to 35% reached an actual final stage (drying rate tending to a plateau), corresponding to the transport of the strongly bounded water still present in food matrix. It is also worthwhile comparing the predictions provided by the present approach with those ones given by a transport model not accounting for shrinkage (Curcio, 2010). A slower decrease of food average moisture content was noticed when shrinkage was accounted for. The observed differences were particularly significant when drying was performed in drastic conditions and almost irrelevant when the exploited process conditions were milder (data not shown). The decrease in time of food dimensions did actually result, due to its definition, in an increase of $\bar{X}_b$. However, the diminution of both L and D also affected the heat and mass transfer rates during either the constant rate period (the transfer rates decreased when food dimensions were smaller) or the falling rate period (a reduc-

tion of transport resistances to heat and mass transfer within food was achieved with a smaller, and partially dehydrated, sample).

4.1. Model validation

In addition to the numerical simulations described in the previous section, some others were performed with the specific aim of verifying the model validity, by checking the agreement with a set of experimental results collected in some of the conditions described in Section 3. Fig. 9 showed the model predictions and the corresponding experimental points expressing the time evolution of potato average moisture content as a function of sample initial diameter. Figs. 10a and 10b showed the model predictions and the corresponding experimental points expressing the time evolution of both L and D at two different inlet temperatures, respectively. In all the considered cases, the agreement was rather good over a time span of about 5 h. It is worthwhile remarking that, as far as food dimensions were concerned, the maximum observed

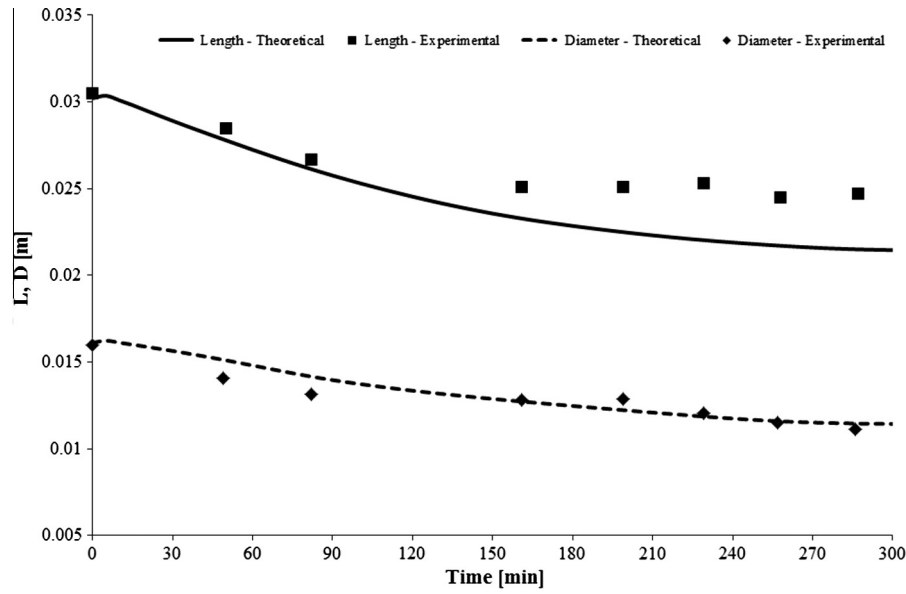


Fig. 10b. Model validation – length and diameter vs. time ($T_{air} = 343$ K, $U_r = 40\%$; $v_0 = 2.8$ m/s; $L_0 = 0.03$ m; $D_0 = 0.016$).

deviations were equal to about 10%; most of the experimental points, however, were remarkably reproduced by the model. Also in the case of $\bar{X}_b$, a very good agreement was observed when the initial diameter was equal to 0.01 m. Some discrepancies, instead, were noticed at intermediate times when a larger initial diameter was considered.

5. Conclusions

A general approach was presented to predict the extent of food shrinkage occurring during convective drying. The obtained results confirmed that it is crucial to develop a model capable of predicting the actual variation of food shape and dimensions. It was indeed proved that deformation strongly influenced the transport phenomena and could not be neglected when a comprehensive food drying model has to be formulated. The proposed approach allowed calculating the time evolution of many different variables, which could be used to ascertain either food safety (microbial inactivation kinetics depend on the local values of temperature and water activity) or its quality (the stresses distribution can be used as a relevant quality index). Simulations results were found to be in good agreement with experimental data in terms of time evolutions of food dimensions and moisture content on a dry basis.

The model, however, could be improved by a more precise description of the actual transport phenomena occurring within the food sample. In particular, it is intended to drop out some of the assumptions used in this paper and regarding the negligibility of both gas pressure gradients and convective heat transfer, as well as the consideration of food sample as a fictitious continuum. In addition, it is proposed to consider a different relationship between free shrinkage strain and water concentration variation, in order to verify if model predictions reliability may be even increased.

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Appendix A. Energy and mass balance equations in the food

Energy balance

$$\rho_s C_p \partial T / \partial t + \nabla \cdot (-k_{eff} \nabla T) + \lambda \cdot \dot{I} = 0 \quad (A.1)$$

Conservation of liquid water

$$\partial C_w / \partial t + \nabla \cdot (-D_w \nabla C_w) + \dot{I} = 0 \quad (A.2)$$

Conservation of water vapor

$$\partial C_v / \partial t + \nabla \cdot (-D_v \nabla C_v) - \dot{I} = 0 \quad (A.3)$$

Combining (A.1)–(A.3), the following expressions were derived:

$$\partial C_w / \partial t + \partial C_v / \partial t + \nabla \cdot (-D_w \nabla C_w) + \nabla \cdot (-D_v \nabla C_v) = 0 \quad (A.4)$$

$$\rho_s C_p \partial T / \partial t + \nabla \cdot (-k_{eff} \nabla T) + \lambda \cdot (-\partial C_w / \partial t - \nabla \cdot (-D_w \nabla C_w)) = 0 \quad (A.5)$$

The vapor pressure of water, p_v , necessary to express the vapor water concentration, depended on potato temperature and moisture content.

$$\ln \frac{p_v}{p_{vs}(T)} = -0.0267 X_b^{-1.656} + 0.0107 \exp(-1.287 X_b) X_b^{1.513} \ln(p_{vs}(T)) \quad (A.6)$$

The physical and the transport properties comparing in Eqs. (A.1)–(A.3) were expressed in terms of the local values of temperature and moisture content (Curcio, 2010).

Appendix B. Transport equations for the drying air

Continuity Eq. (B.1), momentum balance (B.2) and transport equations for k (B.3) and ω (B.4)

$$\partial \rho_a / \partial t + \nabla \cdot \rho_a \underline{u} = 0 \quad (B.1)$$

$$\begin{aligned} \rho_a \partial \underline{u} / \partial t + \rho_a \underline{u} \cdot \nabla \underline{u} + \nabla \cdot (\rho_a \underline{u}' \otimes \underline{u}') \\ = -\nabla p + \nabla \cdot [\eta_a (\nabla \underline{u} + (\nabla \underline{u})^T)] \end{aligned} \quad (B.2)$$

$$\rho_a \partial k / \partial t + \rho_a \underline{u} \cdot \nabla k = \nabla \cdot [(\eta_a + \sigma_k \eta_t)(\nabla k)] + \frac{\eta_t}{2} (\underline{\nabla u} + (\underline{\nabla u})^T)^2 - \beta_k \rho_a k \omega \quad (\text{B.3})$$

$$\rho_a \partial \omega / \partial t + \rho_a \underline{u} \cdot \nabla \omega = \nabla \cdot [(\eta_a + \sigma_\omega \eta_t)(\nabla \omega)] + (\alpha \omega / 2k) \eta_t (\underline{\nabla u} + (\underline{\nabla u})^T)^2 - \beta_\omega \rho_a \omega^2 \quad (\text{B.4})$$

Energy balance:

$$\rho_a C_{pa} \partial T_2 / \partial t - \nabla \cdot (k_a \nabla T_2) + \rho_a C_{pa} \underline{u} \cdot \nabla T_2 = 0 \quad (\text{B.5})$$

Mass balance

$$\partial C_2 / \partial t + \nabla \cdot (-D_a \nabla C_2) + \underline{u} \cdot \nabla C_2 = 0 \quad (\text{B.6})$$

Appendix C. Initial and boundary conditions used to perform the numerical simulations

Initial conditions (C.1):

	Variable	Value
Food	Temperature (T_0)	283 K
	Moisture content on a dry basis (X_{b0})	4.06
	Length (L_0)	0.03 m
	Diameter (D_0)	0.016 m
Air	Temperature (T_{20})	298 K
	Relative humidity (U_{r0})	50%
	Velocity in axial and in radial directions (u_{z0}, u_{r0})	0 m/s
	Pressure in the drying chamber (p_0)	101,325 Pa

Boundary conditions for the transport model equations (C.2):

#: Temperature and concentration profiles were confined to two very thin regions, which developed close to the food-air interface (the validity of such an assumption was already proved by Curcio et al., 2008).

References

- Aregawi, W.A., Defraeye, T., Verboven, P., Herremans, E., De Roeck, G., Nicolai, B.M., 2012. Modeling of coupled water transport and large deformation during dehydration of apple tissue. *Food Bioprocess Technol.* <http://dx.doi.org/10.1007/s11947-012-0862-1>.
- Aversa, M., Curcio, S., Calabrò, V., Iorio, G., 2012. Experimental evaluation of quality parameters during drying of carrot samples. *Food Bioprocess Technol.* 5, 118–129.
- Benboudjemaa, F., Meftah, F., Torrenti, J.M., 2005. Interaction between drying, shrinkage, creep and cracking phenomena in concrete. *Eng. Struct.* 27, 239–250.
- Bernstein, A., Noreña, C.P.Z., 2013. Study of thermodynamic, structural, and quality properties of yacon (*Smallanthus sonchifolius*) during drying. *Food Bioprocess Technol.* <http://dx.doi.org/10.1007/s11947-012-1027-y>.
- Chen, X.D., 2007. Moisture diffusivity in food and biological materials. *Drying Technol.* 25, 1203–1213.
- Curcio, S., 2010. A multiphase model to analyze transport phenomena in food drying processes. *Drying Technol.* 28 (6), 773–785.
- Curcio, S., Aversa, M., Calabrò, V., Iorio, G., 2008. Simulation of food drying: FEM analysis and experimental validation. *J. Food Eng.* 87 (4), 541–553.
- Curcio, S., Saraceno A., Aversa, M. (2012). A theoretical model for the control of color degradation and microbial spoilage occurring in food convective drying. In: Proceedings of the 20th International Congress of Chemical and Process Engineering – CHISA 2012, Prague.
- Datta, A.K., 2007. Porous media approaches to studying simultaneous heat and mass transfer in food processes. I: Problem formulations. *J. Food Eng.* 80, 80–95.
- Dhall, A., Datta, A.K., 2011. Transport in deformable food materials: a poromechanics approach. *Chem. Eng. Sci.* 66, 6482–6497.
- Halder, A., Dhall, A., Datta, A.K., 2007. An improved, easily implementable, porous media based model for deep-fat frying-Part I: Model development and input parameters. *Food Bioprod. Process* 85 (C3), 209–219.
- Halder, A., Dhall, A., Datta, A.K., Black, D.G., Davidson, P.M., Li, J., Zivanovic, S., 2011. A user-friendly general-purpose predictive software package for food safety. *J. Food Eng.* 104, 173–185.
- Hassini, L., Azzouz, S., Peczkalski, R., Belghith, A., 2007. Estimation of potato moisture diffusivity from convective drying kinetics with correction for shrinkage. *J. Food Eng.* 79, 47–56.
- Hernandez, J.A., Pavon, G., Garcia, M.A., 2000. Analytical solution of mass transfer equation considering shrinkage for modeling food-drying kinetics. *J. Food Eng.* 45, 1–10.
- Hirt, C.W., Amsden, A.A., Cook, J.L., 1997. An arbitrary Lagrangian–Eulerian computing method for all flow speeds. *J. Comput. Phys.* 135, 203–216.
- Karathanos, V., 1993. Collapse of structure during drying of celery. *Drying Technol.* 11 (5), 1005–1023.
- Katekawa, M.E., Silva, M.A., 2007. On the influence of glass transition on shrinkage in convective drying of fruits: a case study of banana drying. *Drying Technol.* 25 (10), 1659–1666.
- Kowalski, S.J., 1996. Mathematical modelling of shrinkage during drying. *Drying Technol.* 14 (2), 307–331.
- Kowalski, S.J., Banaszak, J., Rybicki, A., 2010. Plasticity in materials exposed to drying. *Chem. Eng. Sci.* 65, 5105–5116.

	Transport	Expression – description	
Food/air interfaces	Momentum	$\underline{u}/u_\tau = 1/\kappa \ln(\delta_w/l^*) + C^+$	Logarithmic wall function
	Heat	$T = T_2$	Temperature continuity
	Mass	$\underline{n} \cdot (-k_{eff} \nabla T) - \lambda \underline{n} \cdot (-D_w \nabla C_w) = \underline{n} \cdot [(-k_a \nabla T_2) + \rho_a C_{pa} \underline{u} T_2]$	Heat flux continuity
		$C_2 = \gamma_w p_{vs}(T)/RT$ $\underline{n} \cdot (-D_v \nabla C_v) + \underline{n} \cdot (-D_w \nabla C_w) = \underline{n} \cdot [(-D_a \nabla C_2) + \underline{u} C_2]$	Equilibrium Mass flux continuity
Inlet section of the drying cell	Momentum	$u_r = 0, u_z = v_0$	
	Heat	$T_2 = T_{air}$	
	Mass	$C_2 = C_{air}$	Calculated on the basis of U_r values
Drying cell walls	Momentum	$\underline{u}/u_\tau = 1/\kappa \ln(\delta_w/l^*) + C^+$	Logarithmic wall function
	Heat	$T_2 = T_{air}$	(#)
	Mass	$C_2 = C_{air}$	(#)
Outlet section of the drying cell	Momentum	$p = p_{atm}$	
	Heat	$\underline{n} \cdot (-k_a \nabla T_2) = 0$	Convection prevailing over conduction
	Mass	$\underline{n} \cdot (-D_a \nabla C_2) = 0$	Convection prevailing over diffusion

- Kowalski, S.J., Mielniczuk, B., 2006. Drying-induced stresses in macaroni dough. *Drying Technol.* 24 (9), 1093–1099.
- Kowalski, S.J., Rajewska, K., 2002. Drying-induced stresses in elastic and viscoelastic saturated materials. *Chem. Eng. Sci.* 57, 3883–3892.
- Márquez, C.A., De Michelis, A., 2011. Comparison of drying kinetics for small fruits with and without particle shrinkage considerations. *Food Bioprocess Technol.* 4 (7), 1212–1218.
- Mayor, L., Sereno, A.M., 2004. Modelling shrinkage during convective drying of food materials: a review. *J. Food Eng.* 61 (3), 373–386.
- Mihoubi, D., Bellagi, A., 2009. Drying-induced stresses during convective and combined microwave and convective drying of saturated porous media. *Drying Technol.* 27 (7), 851–856.
- Ni, H., Datta, A.K., Torrance, K.E., 1999. Moisture transport in intensive microwave heating of wet materials: a multiphase porous media model. *Int. J. Heat Mass Transf.* 42, 1501–1512.
- Niamnuy, C., Devahastin, S., Soponronnarit, S., Raghavan, G.S.V., 2008. Modeling coupled transport phenomena and mechanical deformation of shrimp during drying in a jet spouted bed dryer. *Chem. Eng. Sci.* 63, 5503–5512.
- Panyawong, S., Devahastin, S., 2007. Determination of deformation of a food product undergoing different drying methods and conditions via evolution of a shape factor. *J. Food Eng.* 78, 151–161.
- Rahman, M.S., 2001. Toward prediction of porosity in foods during drying: a brief review. *Drying Technol.* 19 (1), 1–13.
- Ramos, I.N., Silva, C.L.M., Sereno, A.M., Aguilera, J.M., 2004. Quantification of microstructural changes during first stage air drying of grape tissue. *J. Food Eng.* 62, 159–164.
- Ratti, C., 1994. Shrinkage during drying of foodstuffs. *J. Food Eng.* 23 (1), 91–105.
- Shishegarha, F., Makhlof, J., Ratti, C., 2002. Freeze-drying characteristics of strawberries. *Drying Technol.* 20 (1), 131–145.
- Troncoso, E., Pedreschi, F., 2007. Modeling of textural changes during drying of potato slices. *J. Food Eng.* 82, 577–584.
- Valdramidis, V.P., Geeraerd, A.H., Gaze, J.E., Kondjoyan, A., Boyd, A.R., Shaw, H.L., Van Impe, J.F., 2006. Quantitative description of *Listeria monocytogenes* inactivation kinetics with temperature and water activity as the influencing factors; model prediction and methodological validation of dynamic data. *J. Food Eng.* 76, 79–88.
- Wilcox, D.C., 2004. *Turbulence Modeling for CFD*, second ed. DCW Industries Inc.
- Yadollahinia, A., Jahangiri, M., 2009. Shrinkage of potato slice during drying. *J. Food Eng.* 94 (1), 52–58.
- Yang, H., Sakai, N., Watanabe, M., 2001. Drying model with non-isotropic shrinkage deformation undergoing simultaneous heat and mass transfer. *Drying Technol.* 19 (7), 1441–1460.
- Zhang, J., Datta, A.K., 2004. Some considerations in modeling of moisture transport in heating of hygroscopic materials. *Drying Technol.* 22 (8), 1983–2008.
- Zhang, J., Datta, A.K., Mukherjee, S., 2005. Transport processes and large deformation during baking of bread. *AIChE J.* 51 (9), 2569–2580.